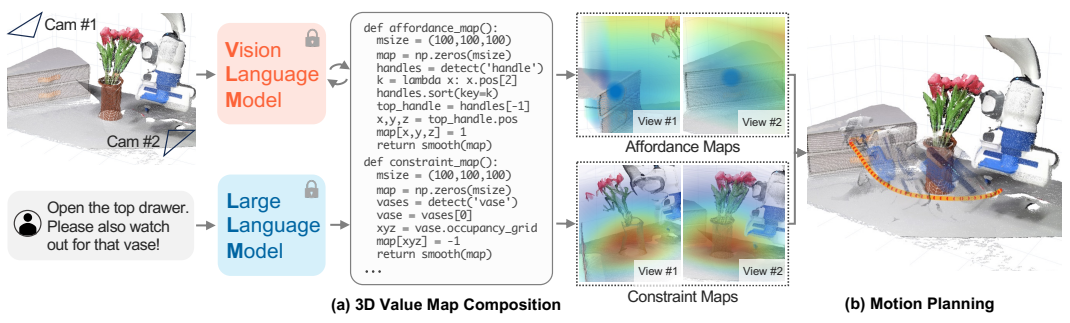

Cam #1
Cam #2
Vision Language Model
Open the top drawer. Please also watch out for that vase!
Large Language Model
def affordance_map():
msize = (100,100,100)
map = np.zeros(msize)
handles = detect('handle')
k = lambda x: x.pos[2]
handles.sort(key=k)
top_handle = handles[-1]
x,y,z = top_handle.pos
map[x,y,z] = 1
return smooth(map)
def constraint_map():
msize = (100,100,100)
map = np.zeros(msize)
vases = detect('vase')
vase = vases[0]
xyz = vase.occupancy_grid
map[xyz] = -1
return smooth(map)
...
View #1
View #2
Affordance Maps
View #1
View #2
Constraint Maps
(a) 3D Value Map Composition
(b) Motion Planning